Supplementary material for: Fine-scale locomotor actuation leaves detectable pressure signatures in fish wakes

S1. Sample accounting

The 40 videos, each containing a different fish, were segmented into 493 samples in which the fish was visible. The following table reports all preprocessing categories and whether samples were retained for two-dimensional simulation and pressure comparison. Filtering was completed before simulation and classifier analyses.

Table S1. Accounting of segmented and retained video samples.

Category	Segmented	Retained	Not retained
Successfully processed	375	367	8
Heavy temporal cropping required	20	17	3
Poor segmentation	11	6	5
No detection in some frames	26	0	26
Fish not fully submerged / 3D interaction	18	0	18
Fish did not pass in front of sensors	26	0	26
Circular segmentation error, tail touching nose	16	13	3
External interference, flake in water	1	0	1
Total	**493**	**403**	**90**

S2. Assessment of fish-identity dependence among retained pressure segments

Because multiple retained pressure segments could originate from one recording, within-fish dependence was evaluated to determine whether particular individuals might dominate segment-level patterns. Each of the 40 fish appeared in one video, so video identity was equivalent to fish identity. The measured-pressure diagnostic was restricted to videos with at least five retained pressure segments.

For this diagnostic, measured pressure segments were transformed with the same preprocessing later used for pressure classification and described in the main article: segments were resampled to a common length, mean-centred and amplitude-normalised. A SVD basis was then computed from the retained measured-pressure matrix alone, and samples were represented by the coefficients of the modes explaining 99% of measured-signal variance. Video-level dependence was quantified in two complementary ways. First, the one-way intraclass correlation coefficient, ICC(1), was computed separately for each retained measured-pressure SVD coefficient. For SVD mode k, the coefficient of segment j from video i was modelled as

$$c_{ij,k} = \mu_k + v_{i,k} + \epsilon_{ij,k}, \tag{1}$$

where $c_{ij,k}$ is the SVD coefficient for segment j from video i on mode k, μ_k is the mean coefficient for mode k computed over all segments, $v_{i,k}$ is the video-level random effect and $\epsilon_{ij,k}$ is the residual segment-level variation. The corresponding per-mode intraclass correlation was

$$ICC_k = \frac{\sigma^2_{\text{video},k}}{\sigma^2_{\text{video},k} + \sigma^2_{\text{extract},k}}, \tag{2}$$

where $\sigma^2_{\text{video},k}$ is the variance component attributable to video identity and $\sigma^2_{\text{extract},k}$ is the residual variance among retained pressure extracts within videos. Values close to zero indicate little video-level clustering, whereas larger values indicate stronger dependence among samples from the same video. Because the pressure representation was multivariate, per-mode ICC values were summarised using the variance-weighted mean

$$\overline{ICC} = \frac{\sum_k \lambda_k ICC_k}{\sum_k \lambda_k}, \tag{3}$$

where λ_k is the measured-pressure variance explained by SVD mode k.

Second, pairwise Euclidean distances in standardised measured-SVD coefficient space were compared between same-video and different-video segment pairs. The ratio between the median same-video and different-video distances provided a descriptive measure of their similarity.

The measured-pressure video-identity analysis retained 426 segments from 23 videos and projected them onto 37 measured-pressure SVD modes explaining 99.1% of variance. The variance-weighted ICC(1) was 0.058, and the median per-mode ICC(1) was 0.019 (range [0; 0.136] with interquartile range [0.005; 0.032]).

In standardised measured-SVD space, the median same-video distance was 7.006 and the median different-video distance was 7.443. The distance ratio was 0.941. Fish identity therefore contributed little to the observed pressure variation, although segments originating from the same source video should not be interpreted as independent biological replicates.

S3. Dataset composition and dimensionality reduction

The following table reports the number of retained pressure segments and the size of the reduced bases used for the pressure-signal classifier. Segments were removed if they were too short or numerically divergent. An equal number of fish and cylinder pressure segments was retained to balance classes.

Table S2. Dataset composition and dimensionality reduction for the four simulation configurations. Raw used original contours and y-only boundary forcing. Size-constrained applied the first-frame fish profile along the reconstructed spine and retained y-only forcing. X-deformation used original contours with both x and y boundary-motion components. Full combined the profile constraint and two-component forcing. Here, x and y are the fixed long and transverse tank axes.

	Raw	Constrained	X-deformation	Full
Samples (fish/cylinder)	163/162	159/161	161/162	162/161
Pressure segments extracted	956/797	796/723	1009/798	702/723
After minimum-length filter	238/236	186/205	200/236	192/205
After divergence filter	238/233	171/202	189/233	186/202
Balanced segments	233/233	171/171	189/189	186/186
Resampled signal length	682	564	580	573
Modes for 75% variance	5	5	23	6
Modes for 95% variance	34	32	92	51
Modes for 99% variance	103	94	153	114

S4. Pressure-signal classifier settings

The pressure-signal classifier was trained after amplitude normalisation, mean-centring, resampling to a common number of points defined by the longest segment in the relevant subset and projection onto SVD modes retaining 75%, 95% or 99% of simulated variance. A separate SVD basis and a separate classifier were fitted for every combination of simulation variant and retained-variance level. Segments were split into training, validation and test subsets in a 70/15/15 ratio. Hyperparameters for a linear classifier with a Mean Square Error loss were selected independently for each combination.

The implementation comprised two consecutive linear layers without an activation function between them. Their composition remained a linear-affine mapping. The reported hidden width parameterised its factorisation but did not introduce a nonlinear decision boundary. This choice maximised interpretability and limited overfitting for the available dataset. Neural-network tooling was retained for optimisation and future extensibility.

Table S3. Hyperparameters selected independently for each model and retained-variance level.

Model	Hidden width	Learning rate	Weight decay	Epochs	Batch size
Raw 75	32	0.0002	0.0002	200	64
Constrained 75	16	0.0006	0.0005	50	16
X-deformation 75	32	0.0002	0.0010	200	64
Full 75	32	0.0010	0.0006	200	64
Raw 95	4	0.0003	0.0009	200	64
Constrained 95	8	0.0009	0.0008	200	64
X-deformation 95	4	0.0010	3.8626×10^{-5}	100	64
Full 95	16	0.0006	0.0002	50	16
Raw 99	4	0.0010	0.0002	200	64
Constrained 99	16	0.0010	0.0009	100	16
X-deformation 99	32	0.0010	0.0008	200	64
Full 99	4	0.0006	0.0006	200	64

S5. Pressure-signal classifier performance

The following accuracies describe how well the classifier separated fish-generated and cylinder-generated simulated pressure signals. These values were used to assess whether the two simulated model classes were distinguishable before measured pressure signals were projected into the same feature spaces.

These values remained moderate rather than near perfect, but this was not unexpected. The fish and cylinder models can generate partially overlapping pressure signatures in some kinematic situations, particularly when the body turns sharply and presents a locally bluff geometry to the sensor. Perfect separability should therefore not be expected from sparse wall-pressure measurements alone.

Multiple simulated pressure segments could originate from the same source video and consequently share the same experimentally measured trajectory. Fish simulations from that video also shared the same image-derived contour sequence. A segment-level split could therefore place intervals derived from the same recording in both fitting and evaluation data, making estimated fish–cylinder separation partly dependent on source-video characteristics. A secondary grouped

Table S4. Classification accuracies for the first-stage pressure-signal classifier.

	Raw	Constrained	X-deformation	Full
Training accuracy (75%)	0.67	0.68	0.75	0.67
Validation accuracy (75%)	0.69	0.65	0.81	0.82
Test accuracy (75%)	0.74	0.63	0.84	0.61
Training accuracy (95%)	0.70	0.74	0.74	0.71
Validation accuracy (95%)	0.70	0.65	0.77	0.73
Test accuracy (95%)	0.76	0.65	0.84	0.57
Training accuracy (99%)	0.78	0.75	0.80	0.75
Validation accuracy (99%)	0.64	0.65	0.84	0.66
Test accuracy (99%)	0.67	0.60	0.79	0.55

robustness check tested separation when entire source videos were excluded from centroid fitting.

The check used the x-deformation variant and the existing pressure-mode basis retaining 99% of simulated variance. After the existing filters and class balancing, it included 189 fish and 189 cylinder segments from 27 source videos. For each of 100 repetitions, seven complete videos were randomly held out. Fish and cylinder centroids were calculated in pressure-mode coefficient space from the remaining videos, and each held-out segment was assigned to its nearest centroid. The resulting mean accuracy was 0.747 (SD 0.056, range 0.600–0.911). This was not a retraining of the primary classifier. The SVD basis had also been computed previously from all eligible segments, so the analysis tested robustness to source-video grouping at the centroid-classification stage rather than providing fully nested grouped validation.

S6. Details of contour-size consistency

The contour extracted from each video frame was treated independently in the original pipeline. Its projected size could therefore fluctuate because of segmentation noise, natural posture variation or out-of-plane motion such as body roll. A contour-size consistency constraint was introduced to test whether regularising these fluctuations improved agreement with measured pressure. The fish centreline, termed the spine, was first estimated from the raw contour by partitioning the body into upper and lower regions of equal area. The spine was then extrapolated at the snout and tail tips. Finally, a fish profile defined from the first video frame was placed along the spine and optimised against the raw detection by minimising the intersection-over-union mismatch.

The size-constrained sequence therefore retained bending of the reconstructed spine while imposing the first-frame profile and contour length. Subsequent simulation stages were unchanged.

S7. Details of tank-axis deformation in the CFD twin

The simulation used fixed tank coordinates: x followed the tank's long axis and y its transverse axis. Fish trajectories were predominantly oriented along x, so x was approximately streamwise for most samples and lateral undulation was expressed mainly along y. These terms do not denote instantaneous fish-centred axes.

LilyPad describes interaction between the moving body and fluid through functions for body distance and temporal and spatial boundary variation. In the default implementation used as the starting point, moving-boundary forcing was hard-coded to use only the y component. In the raw model, imported contours still placed each boundary point at its correct x, y position in every frame, but local x displacement between frames was not fully included in the boundary velocity communicated to the fluid. The y-only implementation was a deliberate first approximation because idealised fish swimming is often represented as progression along the swimming direction with deformation dominated by lateral undulation. Most observed trajectories were also aligned approximately with the tank's x axis. Real trajectories and body motion could depart from this approximation during turns or when the tail tip moved towards or away from the head.

In the updated implementation, boundary velocity was computed directly from contour-point displacement between consecutive time steps. Both x and y components were used, with local averaging over neighbouring boundary nodes. The contour input was unchanged. This allowed apparent longitudinal redistribution, local extension and compression of the projected contour and other secondary in-plane motions to contribute to the simulated flow.

Detected vortex cores were assigned to the nearest of the three analysed sensor coordinates and retained within 32 grid cells (32 mm). This proximity threshold was chosen heuristically by visual inspection to exclude vortices remote from a sensor and was not calibrated as a biological sensing radius. LilyPad defines a three-grid-cell numerical bounding region around the body. The baseline minimum vortex-to-body distance was doubled to six grid cells to avoid extracting structures at its edge. A pressure segment was a contiguous interval during which at least one retained vortex assigned to a sensor was visible. Sensitivity to body-distance thresholds of 3 and 12 grid cells was evaluated separately.

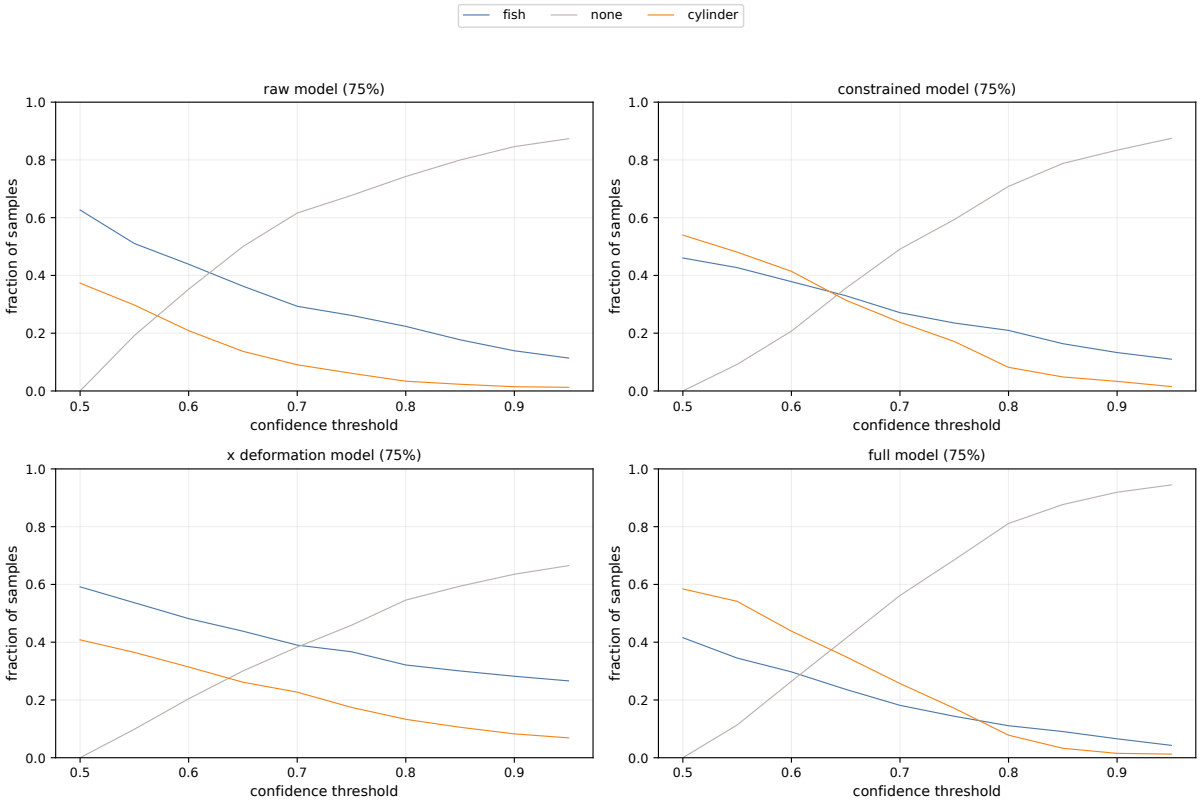
S8. Vortex descriptors used for interpretation

The second-stage classifier used 17 physically interpretable vortex descriptors: maximum area, mean area, displacement, path length, duration in frames, duration ratio of the vortex life over the full segment duration, mean distance to the body, mean distance to the sensor, maximum speed, mean speed, maximum radius, mean radius, maximum core Q-criterion value, mean core Q-criterion value, mean absolute core Q-criterion value, mean core x-position and mean core y-position.

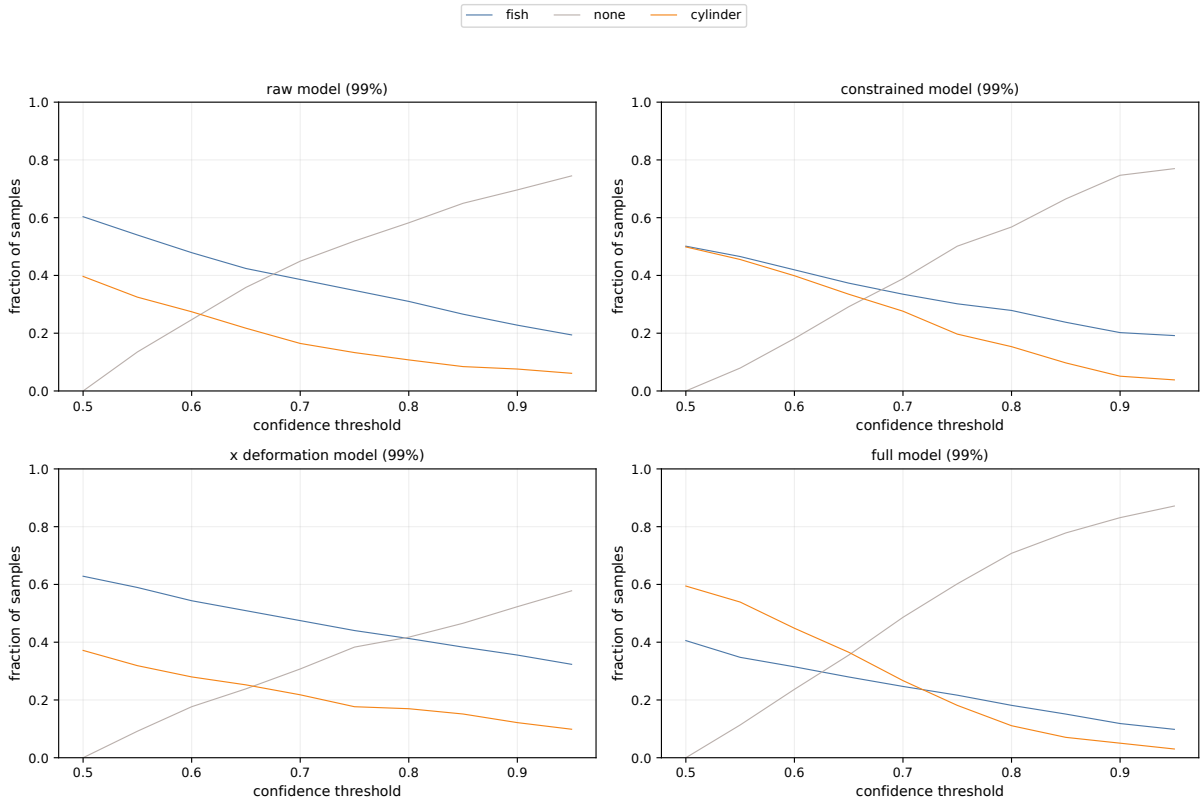
Vortices were retained only if they were present for at least 10% of the duration of the linked pressure segment. The two classes were balanced, each feature was normalised and mean-centred, and the same 70/15/15 data split strategy was used again.

Because the x-deformation model with 95% retained variance provided the clearest separation in the first-stage analysis, this subset was selected for vortex-based interpretation. At a 99% confidence threshold, 63 fish-labelled and 63 cylinder-labelled vortices were available for the second-stage training set. The final hyperparameters were a learning rate of 0.0069, weight decay of 2×10^{-6}, 75 training epochs and a batch size of 8. On the test set, the second classifier achieved an F1 score of approximately 0.80 for both classes.

S9. Results for 75% and 99%



Supplementary Figure 1. Fractions of measured pressure-signal segments classified as fish-like, cylinder-like or undetermined when 75% of variance was retained. Across most configurations, measured signals were more frequently associated with the fish-shaped model than with the cylinder surrogate.



Supplementary Figure 2. Fractions of measured pressure-signal segments classified as fish-like, cylinder-like or undetermined when 99% of variance was retained. Across most configurations, measured signals were more frequently associated with the fish-shaped model than with the cylinder surrogate.